10 Striking Mechanisms

10.1 The alarm watch

In an alarm watch, which may also be used as a memory aid, the mechanism which releases the alarm is identical to that in an alarm clock. Part of the movement is totally conventional - simple, calendar or self-winding - with a stem for winding and adjusting the hands.

We shall study only the **alarm mechanism** here, which has its own stem for winding the mechanism and setting the alarm time.

fig. 10-1 : An alarm watch.

10.1.1 The alarm mechanism

This mechanism comprises the alarm winding stem *(1)* with its winding pinion *(2)*, transmission wheels *(3)*, a ratchet *(4)* and the alarm barrel *(5)* *(fig. 10-2)*.

The alarm barrel gears directly with the strike wheel and pinion *(6)*. The wheel has 11 triangular teeth with rounded tips. The strike wheel is in fact an escape wheel which is controlled by the alarm pallets *(7)* to which a hammer *(8)* is fixed. The hammer strikes a pin fixed at the back of the case (generally at the centre of the movement) or a gong. The frequency of the strike depends on the torque of the mainspring and the mass of the hammer.

The release wheel *(10)* pivots above the hour wheel *(9)* (dial side). The release lever *(11)*, with its own spring *(12)*, presses the hour wheel against the release wheel.

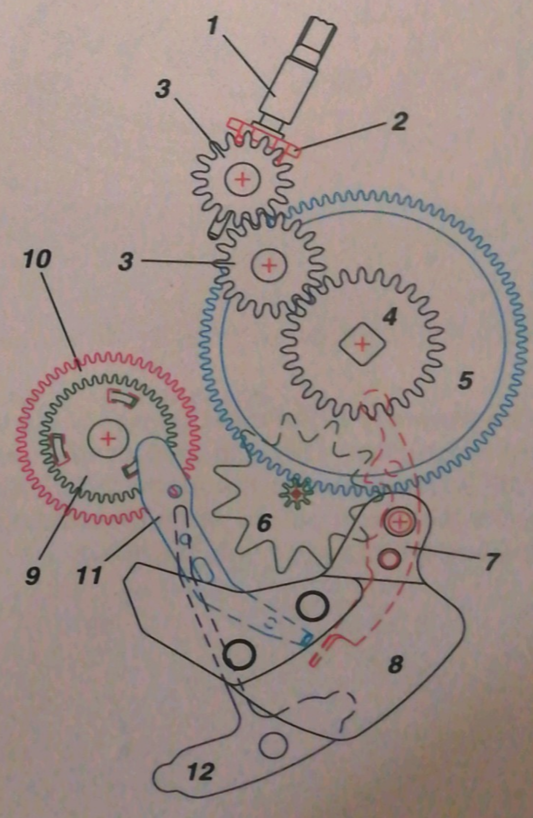

fig. 10-2

10.1.2 The release of the alarm mechanism

In the hand-setting position the alarm winding stem turns the release wheel through 3 intermediate wheels ***(13)*** *(fig. 10-3)*.

The pipe of the release wheel has 4 notches which correspond to those in the alarm disc at the centre of the dial. This disc has an index that can be adjusted to the desired release time *(see fig. 10-1)*.

The plate of the release wheel has three curved slots on three different radii *(fig. 10-4)*.

The hour wheel has three angled catches in its disc *(figs. 10-5 and 10-6)* and is pressed against the release wheel by the release lever ***(11)*** through its spring ***(12)*** *(figs. 10-3 and 10-7)*.

As it turns, and at the time set for the alarm to go off, the three catches on the hour wheel are aligned with, and slip into, the slots in the release wheel against which the hour wheel is pressed *(fig. 10-8)*.

The lever tilts and releases the alarm pallets ***(7)***. The alarm will continue to sound until the alarm spring ceases to provide power.

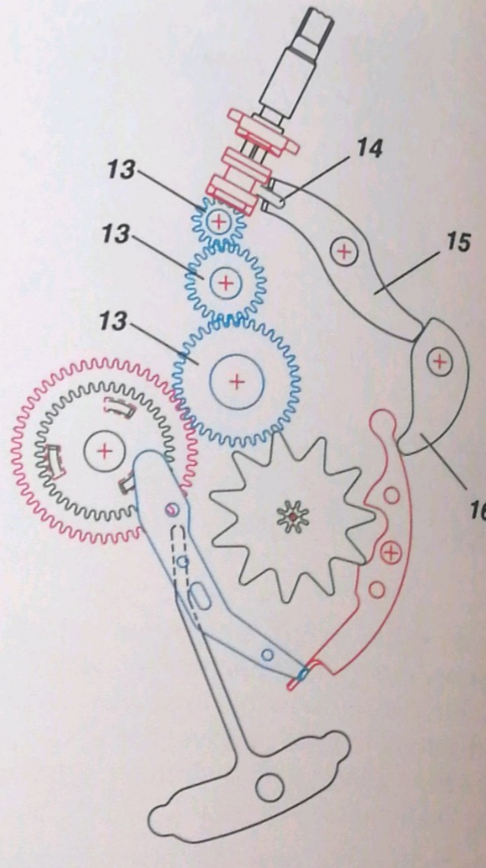

fig. 10-3

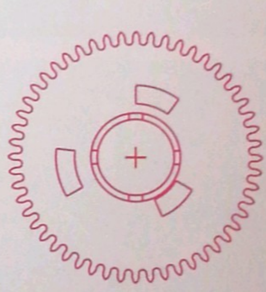

fig. 10-4

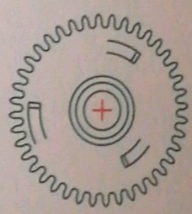

fig. 10-5

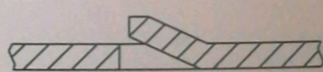

fig. 10-6 : Cross section of a catch.

10.1.3 The alarm lock

The extremity of the rocking lever of the alarm winding mechanism ***(14)*** is in contact with the slide lever ***(15)*** which activates the alarm lock ***(16)***. The latter locks the alarm pallets when the winding stem is in the hand-setting position *(fig. 10-3)*.

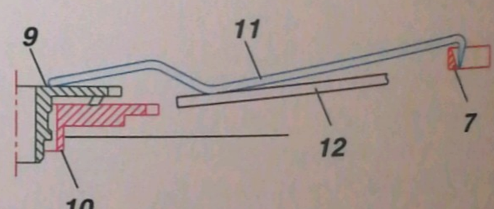

fig. 10-7

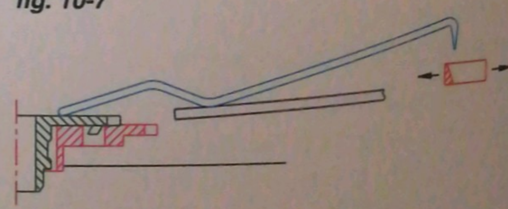

fig. 10-8

10.2 The minute-repeater mechanism

This mechanism is highly complex and very complicated to adjust perfectly. Specialised literature must be studied in order to find out all about the minute-repeater mechanism.

The role of the mechanism

A minute-repeater mechanism is used to tell the time to the nearest minute on demand, which is indicated by the sound of two hammers hitting two gongs of different pitches *(fig. 10-9)*.

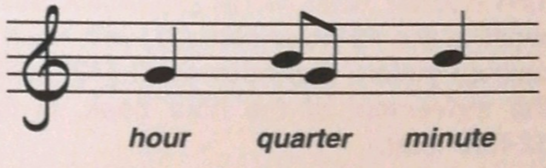

fig. 10-9

10.2.1 The hammers, gongs and hammer trips

The **hammers**, made out of hardened steel, pivot on staffs. They are located on the bridge side, one on each side of the repeating barrel. The larger hammer ***(1)*** (or hour hammer) hits a low-pitched **gong *(2)***, while the smaller hammer ***(3)*** (or minute hammer) hits a high-pitched gong ***(4)*** *(figs. 10-10 and 10-11)*.

A pin is fixed to each hammer at the side of the staff. This pin passes through an elongated hole in the plate and serves as the connection to the **pallet or hammer trip** which lifts the hammer. Two return springs ***(5 and 6)*** act on the two pins to allow the hammers to strike the gongs.

In their position of rest the hammers do not touch the gongs thanks to two countersprings ***(7 and 8)*** which are slightly flexible; in this way the gongs can vibrate freely *(fig. 10-11)*.

The gongs are thin rods with a circular cross section, made out of hardened steel, and encircle the bridges. They are fixed rigidly on to the plate by an integral block.

The hammer trips are small levers with an integral tooth; they are lifted by the striking mechanism. Each hammer has two hammer trips.

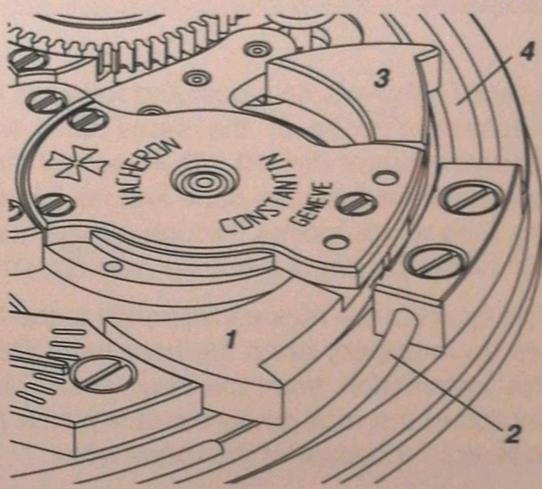

fig. 10-10 : Bridge side.

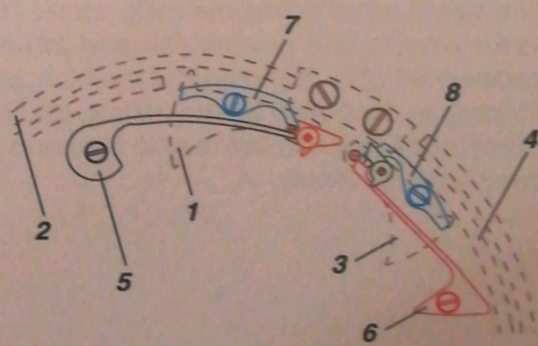

fig. 10-11 : Dial side.

10.2.2 Access to the repeating mechanism "memory"

The watch movement is conventional in design. The **quarter snail *(9)***, which controls the travel of the quarter rack, and the **minute snail *(10)***, which controls the travel of the minute rack, are fixed to the cannon-pinion shank *(fig. 10-12)*.

The quarter snail has a pin which advances one tooth of the **12-hour star *(11)*** with each turn of the cannon pinion. The **hour snail *(12)***, which controls the movement of the hour beak, is fixed to the 12-hour star.

The position of the 12-hour star is controlled by a jumper ***(13)***.

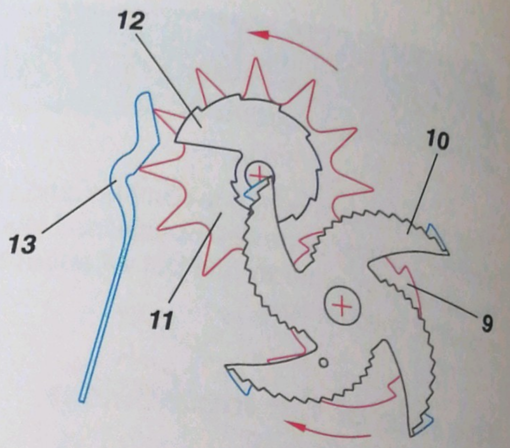

fig. 10-12

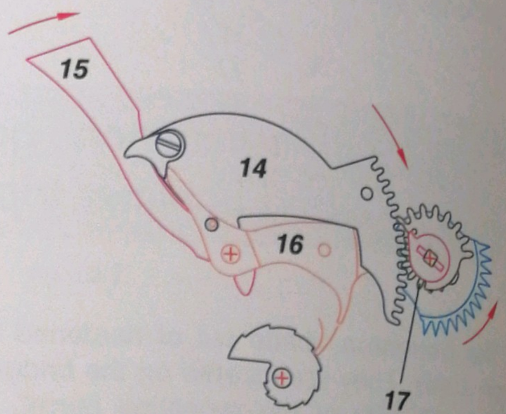

fig. 10-13

10.2.3 The driving force of the repeater mechanism

A rack ***(14)*** pivots on a post and releases the striking mechanism while at the same time winding the repeating barrel *(fig. 10-13)*. An arm of the rack ***(15)*** projects from the case band and fits into the repeating slide which moves through a maximum angle of 25° along the rim of the case *(see fig. 10-28)*.

The travel of the rack corresponds to the hour which will be indicated, since the hour beak ***(16)*** picks up this information from the hour snail. The toothed section of the rack drives the rack pinion ***(17)***, which is fixed to a square of the barrel arbor.

There is a toothed ratchet on the upper rim of the repeating barrel which is retained by a click. This is used to "set up" or pre-tension, the repeating spring.

The barrel arbor drives the main wheel ***(18)***, the two strike wheels and pinions ***(19 and 20)*** and the escape wheel ***(21)***. The speed of the train, and thus the striking, is controlled by the pallets ***(22)*** *(fig. 10-14)*. Sometimes this escapement is replaced by a silent centrifugal governor.

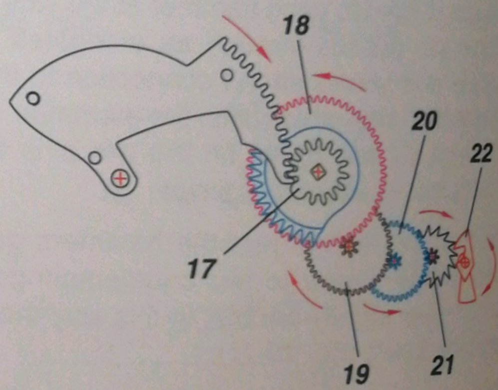

fig. 10-14

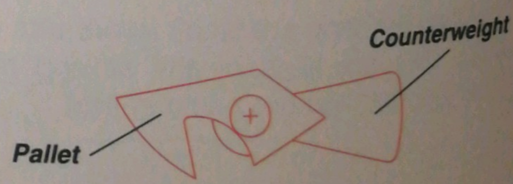

fig. 10-15 : The repeating escapement.

10.2.4 The hour strike

There are 12 pointed teeth on a section of the circumference of the **hour rack** ***(23)***, which is screwed beneath the rack pinion *(fig. 10-16).*

Depending on the travel of the rack, which is restricted by the hour snail, a certain number of teeth on the ratchet will pass ready to activate the large hammer through the hour-hammer trip ***(24)***, which pivots freely on its axis.

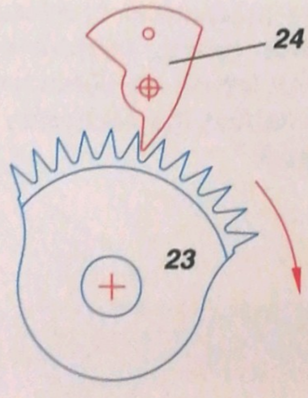

fig. 10-16

10.2.5 The quarter strike

The **quarter rack** ***(25)***, which pivots on a post, is driven by a pinion ***(26)*** which gears with its internal teeth. This pinion moves freely on the repeating barrel arbor and is driven by a finger ***(27)*** fixed to a square of the arbor.

The quarter snail limits the travel of the beak ***(28)*** and determines the position of the quarter rack. There are two groups of three teeth on the rim of the quarter rack; the first group drives one quarter-hammer trip (small hammer) ***(29)*** and the second the other quarter-hammer trip (large hammer) ***(30)***. The spacing of these teeth is designed in such a way that the high and low-pitched tones alternate to give a "ting-tang" for each quarter *(fig. 10-17).*

The minute-rack driving hook ***(31)*** is fixed to the quarter rack *(fig. 10-18).*

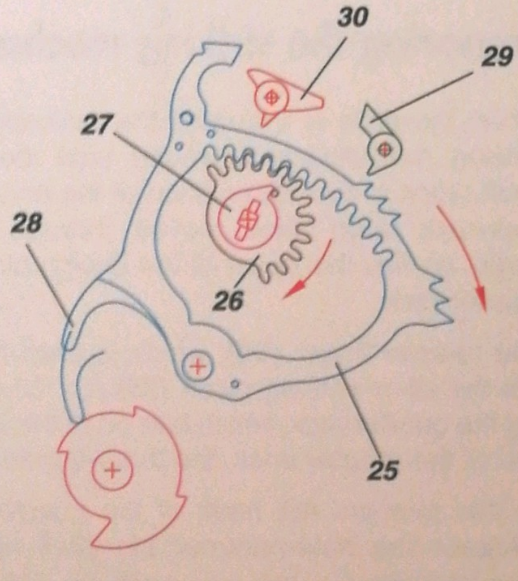

fig. 10-17

10.2.6 The minute strike

The **minute rack** ***(32)*** pivots on the same axis as the quarter rack. There are 14 teeth on the rim of the minute rack, which come into contact with the minute-hammer trip ***(33)*** *(fig. 10-18).*

The minute rack is driven by the minute-rack driving hook, which pivots on the quarter rack and picks up one of the 6 driving teeth, depending on the number of minutes to be struck.

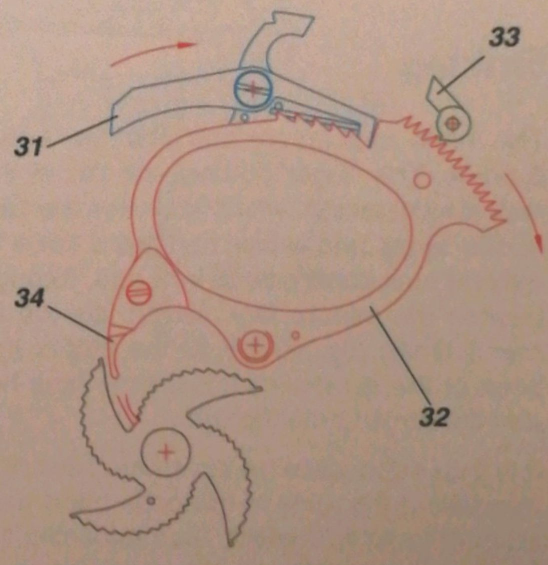

fig. 10-18

When the mechanism is released *(see 10.2.7)* the minute-rack driving hook frees the minute rack, which then travels as determined by its beak ***(34)***, and thus derives the necessary information from the minute snail.

10.2.7 Releasing the racks

Let us suppose that the time shown by the watch is 8.51 and the minute-repeater mechanism is required to indicate this *(fig. 10-19)*.

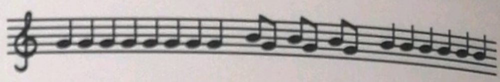

fig. 10-19

Preparing the striking mechanism

When the slide is activated the mainspring for the striking mechanism is wound until the hour-rack beak stops at the 8th position of the hour snail. The hour-rack teeth have moved through the same angle, as has the finger of the driving pinion for the quarter rack.

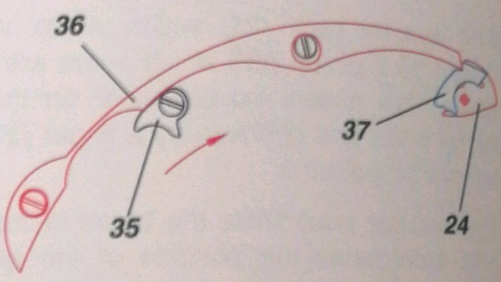

fig. 10-20

The release finger ***(35)***, which is fixed to the rack, lifts the all-or-nothing piece ***(36)*** *(fig. 10-20)*, releasing the quarter rack, which falls on to the lowest position of the quarter snail, the three-quarters position.

At this moment the beak of the quarter rack ***(37)*** releases the hour-hammer trip ***(24)***, allowing the hour to strike.

At the same time the minute-rack driving hook ***(31)*** releases the minute rack ***(32)*** and its beak derives the required information from the minute snail: in this case, the 6th minute *(fig. 10-21)*.

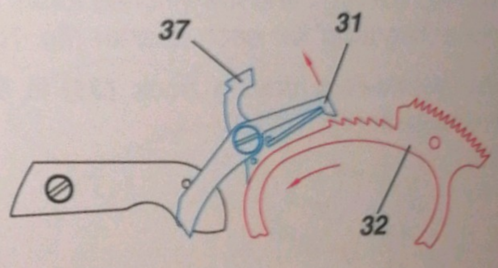

fig. 10-21

Striking

The hour rack turns and lifts the large hammer 8 times. The finger gathers the pin of the quarter-rack driving pinion, which activates the hammer trips of the large and small hammers three times. The quarter rack continues to turn and, through its hook, pushes the minute rack, which lifts the small hammer 6 times *(fig. 10-22)*. At the end of its travel the beak of the quarter rack locks the hour-hammer trip and the striking mechanism.

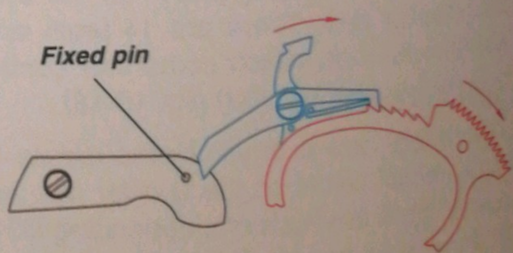

fig. 10-22

It is thus impossible for the striking mechanism to be released if the slide in the case band has not been pushed to its full extent, thus releasing the racks in sequence as explained above.

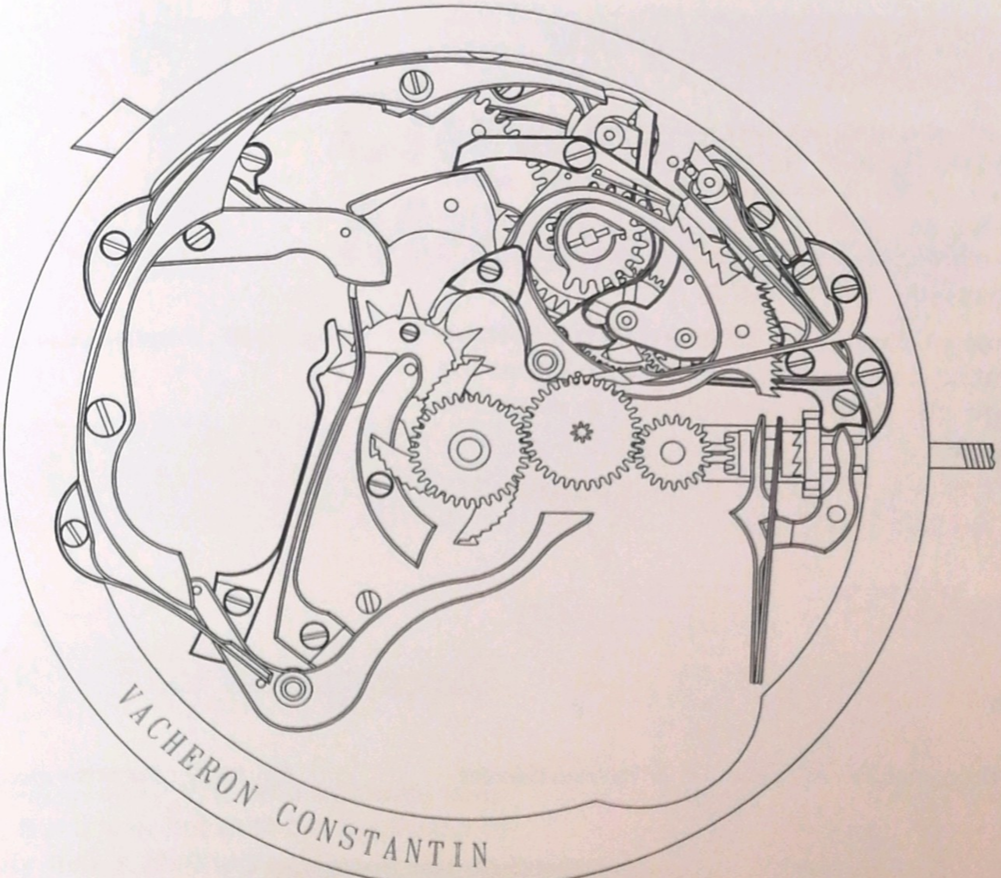

fig. 10-23 : Mechanism in position of rest.

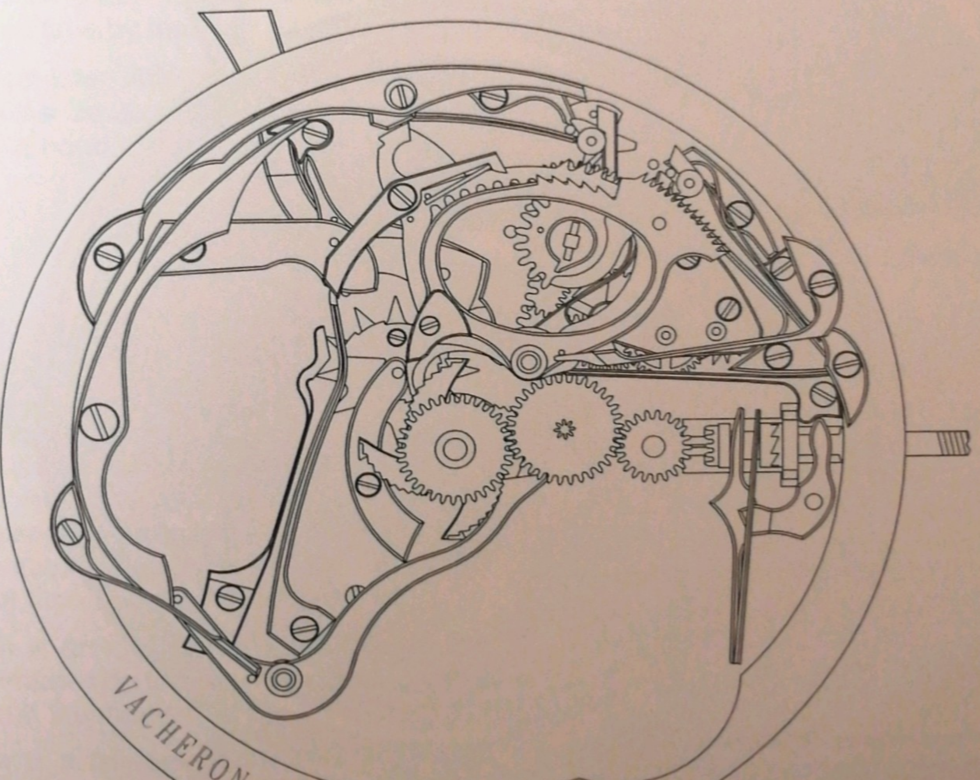

fig. 10-24 : Mechanism fully wound, ready to be released and strike.

10.2.8 The surprise piece

The quarter and minute snails ***(9 and 10)*** are fitted with a plate called a **surprise piece *(38)***, which moves in an arc that is sufficient to temporarily extend the extremities of the snails; it is withdrawn to allow the passage of the fingers.

The purpose of this device is to avoid the mechanism indicating three-quarters or 14 minutes at the instant when the 12-hour star jumps, i.e. when the hour changes.

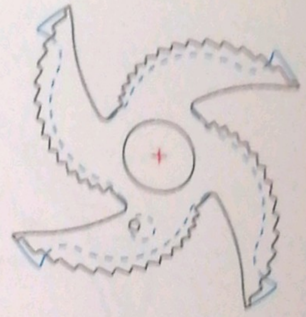

fig. 10-25 : Surpise piece in position of rest.

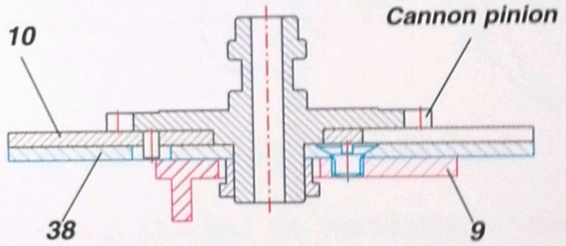

fig. 10-26 : Cross-section of the centre of the mechanism.

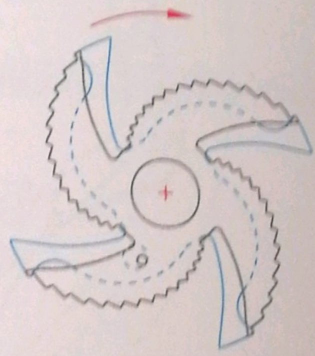

fig. 10-27 : Surprise piece extended.

fig. 10-28 : A Vacheron Constantin watch with a minute-repeater mechanism.